

ST BENEDICT'S

SUBJECT	Mathematics	PAPER	Analytical Geometry
GRADE	11	DATE	28 February 2024
EXAMINER NAME	S McConkey	MARKS	45
TEACHER	MEMO	MODERATORS	T Meredith; M Brown
		DURATION	50 min

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Analytical Geometry Basics	16	
2	Inclination of a line	11	
3	Equations of lines	13	
4	Application question	5	
TOTAL		45	

INSTRUCTIONS:

- This paper consists of 4 questions and ⁹~~10~~ pages. The last page can be used for additional working, if necessary. If this space is used, make sure that you indicate clearly which question is being answered.
- Read the questions carefully.
- Answer all questions asked and answer them in the spaces provided on the question paper.
- You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
- Round off your answers to **one** decimal digit, where necessary.
- All necessary working details must be shown. Answers only, without the relevant calculations will not necessarily be given full marks.
- It is in your interest to write legibly and present your work neatly.

USEFUL FORMULAE:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$M \left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2} \right)$$

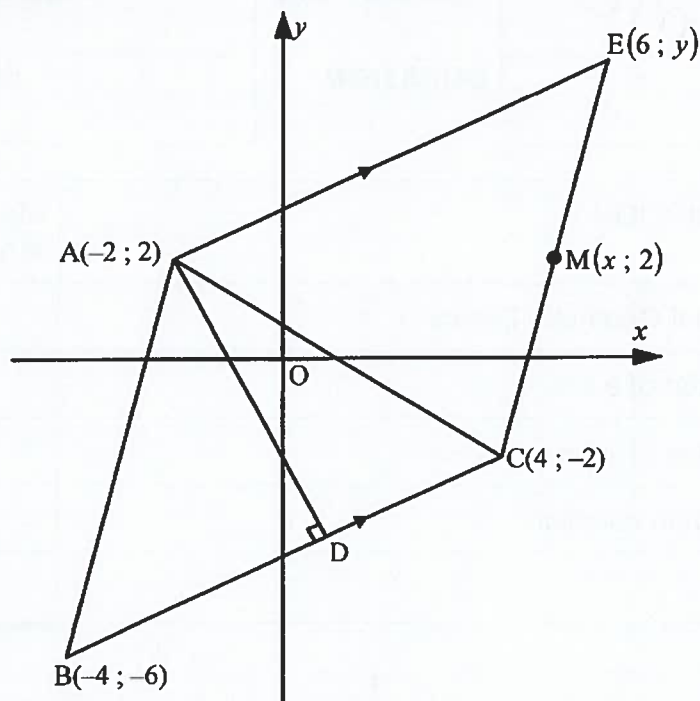
$$m = \tan \theta$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

QUESTION 1**16 MARKS**

In the diagram, A (-2; 2), B (-4; -6), C (4; -2) and E (6; y) are the vertices of a quadrilateral with $AE \parallel BC$. D lies on BC such that $AD \perp BC$, and AC is drawn. M (x; 2) is a point on EC.



- a) Calculate the gradient of BC.

$$m_{BC} = \frac{-6 + 2}{-4 - 4} = \frac{1}{2}$$

answer only
Full marks (2)

- b) If M is the midpoint of EC, determine the values of x and y.

(3)

$$x = \frac{6 + 4}{2}$$

$$= 5$$

$$\frac{y - 2}{2} = 2$$

$$y - 2 = 4$$

$$y = 6$$

(only if $y - 2 = 4$ is shown,
as $y = 6$ given ind)

- c) Calculate the length of BC, leaving your answer in simplest surd form. (2)

$$\begin{aligned}
 dBC &= \sqrt{(4+4)^2 + (-2+6)^2} \quad \checkmark a \\
 &= \sqrt{64 + 16} \\
 &= \sqrt{80} \\
 &= 4\sqrt{5} \quad \checkmark ca \quad \text{must be in surd form}
 \end{aligned}$$

- d) Given that $y = 6$ units, calculate the length of AE, and hence, determine what type of quadrilateral ABCE is. Give a reason for your answer. (3)

$$\begin{aligned}
 dAE &= \sqrt{(6+2)^2 + (6-2)^2} \\
 &= \sqrt{80} \\
 &= 4\sqrt{5} \quad \checkmark a
 \end{aligned}$$

$\therefore (AE = BC \quad \text{and} \quad AE \parallel BC) \quad \checkmark$ or (one pr opp sides $\parallel$ and $=$) explanation
 ABCE is a parallelogram. a

- e) Determine the equation of AD in the form $y = mx + c$. (3)

$$\begin{aligned}
 mBC &= \frac{1}{2} \quad (\text{from a}) \\
 \therefore mAD &= -2 \quad \checkmark ca
 \end{aligned}$$

$$\begin{aligned}
 2 &= -2(-2) + c \\
 &= 4 + c \\
 -2 &= c \\
 \therefore y &= -2x - 2
 \end{aligned}$$

$$\begin{aligned}
 y - 2 &= -2(x + 2) \\
 y &= -2x - 4 + 2 \\
 &= -2x - 2
 \end{aligned}$$

ca $\checkmark$

- f) Determine the area of ΔABC , given $D\left(\frac{4}{5}; -\frac{18}{5}\right)$. (3)

$$d_{AD} = \sqrt{\left(\frac{4}{5} + 2\right)^2 + \left(-\frac{18}{5} - 2\right)^2}$$

$$= \frac{14\sqrt{5}}{5} \checkmark a \approx 6,3$$

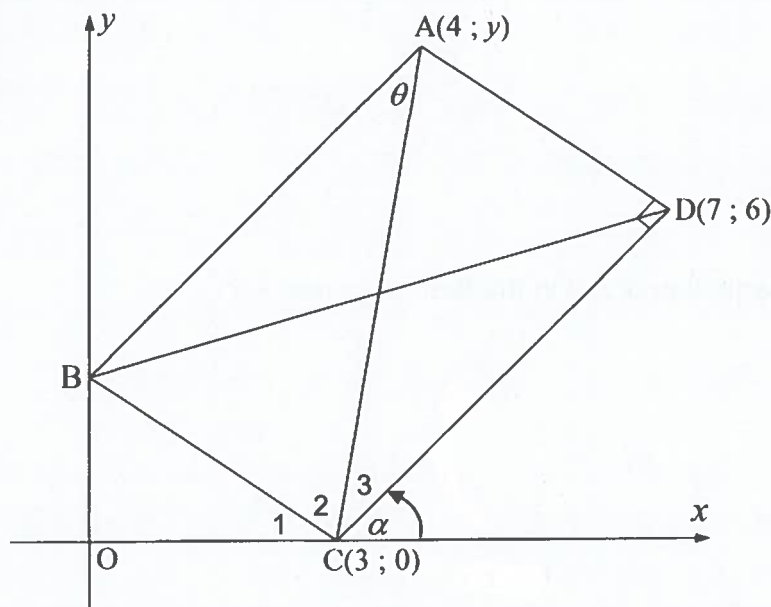
$$\text{Area} = \frac{1}{2} (4\sqrt{5}) \left(\frac{14\sqrt{5}}{5}\right) \checkmark ca. \quad \frac{1}{2} BC \cdot AD$$

$$= 28 \text{ units}^2 \checkmark ca. (\approx 28,2)$$

QUESTION 2

11 MARKS

In the diagram, B is a point on the y-axis. A (4; y), B, C (3; 0) and D (7; 6) are the vertices of **rectangle** ABCD. Diagonals BD and AC are drawn. The inclination of DC is α and $\hat{BAC} = \theta$.



- a) Calculate the size of α . (3)

$$m_{CD} = \frac{6-0}{7-3} = \frac{3}{2} \checkmark a$$

$$\tan \alpha = \frac{3}{2} \checkmark ca$$

$$\alpha = 56,3^\circ \checkmark ca$$

b) Determine the value of y .

(4)

$$m_{AD} = \frac{y-6}{4-7} = \frac{y-6}{-3} \quad \checkmark a$$

(Can use Pythag)

$$m_{DC} = \frac{3}{2}$$

$$\therefore \frac{y-6}{-3} = -\frac{2}{3} \quad \checkmark m$$

ca $\checkmark$ (from a)

(Find equation of AD. Subs A)

$$3y - 18 = 6$$

$$3y = 24$$

$$y = 8$$

$\checkmark ca$ Only if previous step correct

c) Calculate the size of θ , given that $y = 8$.

(4)

$$m_{AC} = \frac{8-0}{4-3} = 8 \quad \checkmark a$$

$$\tan x = 8$$

$$x = 82,9^\circ \quad \checkmark ca$$

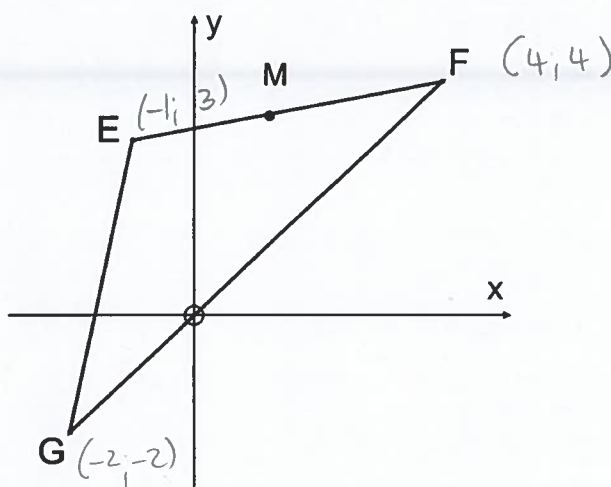
$$\begin{aligned} \hat{C}_3 &= 82,9 - 56,3 \quad \checkmark m \\ &= 26,6^\circ \end{aligned}$$

$$\therefore \theta = 26,6^\circ \quad \checkmark ca \text{ alt \u00c7s } AB \parallel CD$$

QUESTION 3

13 MARKS

In the diagram below, ΔEFG has vertices $E(-1; 3)$, $F(4; 4)$ and $G(-2; -2)$. M is the midpoint of EF .



- a) Determine the coordinates of M .

(2)

$$M \left(\frac{-1+4}{2}, \frac{3+4}{2} \right) = \left(\frac{3}{2}, \frac{7}{2} \right) \quad \checkmark_a \quad \checkmark_a$$

- b) Given that N is the point $\left(\frac{1}{2}; p\right)$, determine the possible value(s) of p given that the length of MN is $\sqrt{10}$ units.

(5)

$$\begin{aligned} d_{MN} &= \sqrt{\left(\frac{1}{2} - \frac{3}{2}\right)^2 + \left(p - \frac{7}{2}\right)^2} \quad \checkmark_{ca} \\ &= \sqrt{1 + p^2 - 7p + \frac{49}{4}} \\ \sqrt{p^2 - 7p + \frac{53}{4}} &= \sqrt{10} \quad \checkmark_m \\ p^2 - 7p + \frac{53}{4} &= 10 \\ 4p^2 - 28p + 13 &= 0 \quad \checkmark_{ca} \\ (2p-1)(2p-13) &= 0 \\ p &= \frac{1}{2} \quad \checkmark_{ca} \quad \text{or} \quad p = \frac{13}{2} \quad \checkmark_{ca} \end{aligned} \quad \left\{ \begin{aligned} \sqrt{\left(p - \frac{7}{2}\right)^2 + 1} &= \sqrt{10} \quad \checkmark_m \\ \left(p - \frac{7}{2}\right)^2 + 1 &= 10 \\ \left(p - \frac{7}{2}\right)^2 &= 9 \\ p - \frac{7}{2} &= \pm 3 \quad \checkmark_{ca} \\ p &= \frac{7}{2} \pm 3 \\ p &= \frac{1}{2} \quad \text{or} \quad \frac{13}{2} \\ \checkmark_{ca} & \quad \checkmark_{ca} \end{aligned} \right.$$

- c) Determine the equation of the line which is perpendicular to EF and passes through point M. (4)

$$m_{EF} = \frac{1}{5} \quad \checkmark a$$

$$\perp m = -5 \quad \checkmark ca \left(\frac{3}{2}, \frac{7}{2} \right)$$

$$y - \frac{7}{2} = -5 \left(x - \frac{3}{2} \right) \quad \checkmark ca \quad \frac{7}{2} = -5 \left(\frac{3}{2} \right) + c$$

$$y = -5x + \frac{15}{2} + \frac{7}{2}$$

$$= -\frac{15}{2} + c$$

$$11 = c$$

$$= -5x + 11$$

$$\checkmark ca \quad \therefore y = -5x + 11$$

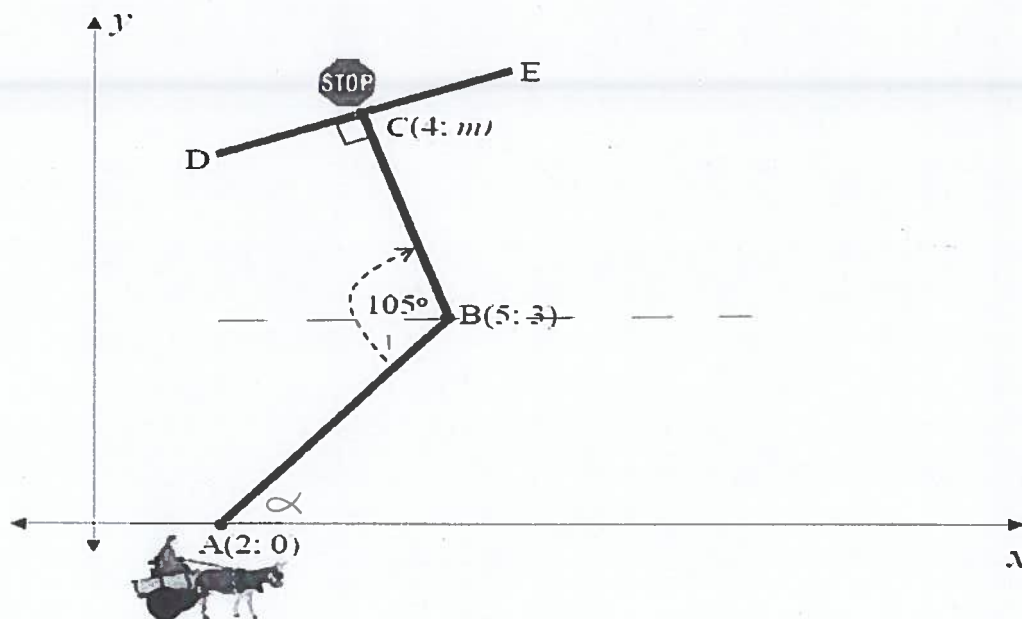
- d) Write down the coordinates of H, if EFGH is a parallelogram. (2)

$$H(-7; -3) \quad \checkmark a \quad \checkmark a$$

QUESTION 4

5 MARKS

A donkey cart travels along a dirt road from A to B to C, with coordinates as shown in the diagram below.



Show that the value of $m = 3 + \sqrt{3}$. Hint: Draw a horizontal line through B.

$$m_{AB} = \frac{3}{2} = 1.5 \quad \checkmark_{ca}$$

$$\tan \alpha = 1$$

$$\alpha = 45^\circ \quad \checkmark_{ca}$$

$$\therefore \hat{B}_1 = 45^\circ$$

$$\therefore \text{inclination of } CB = 120^\circ \quad \checkmark_{ca}$$

$$m = \tan 120^\circ$$

$$= -\sqrt{3} \quad \checkmark_{ca}$$

$$m_{CB} = \frac{m-3}{-1}$$

$$-(m-3) = -\sqrt{3} \quad \checkmark_{ca}$$

$$m-3 = \sqrt{3}$$

$$m = 3 + \sqrt{3}$$